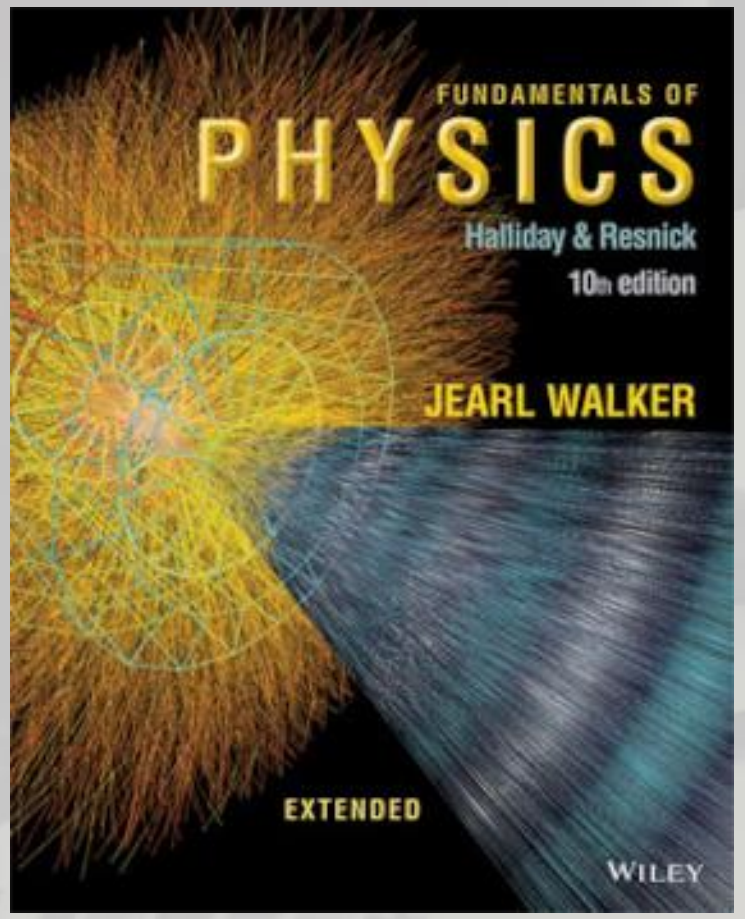
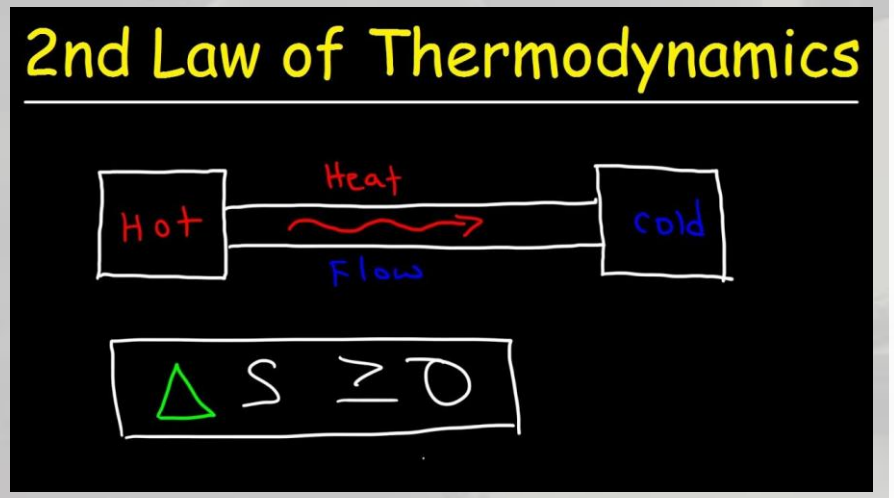
IV. Entropy and the Second Law of Thermodynamics



Outcomes

- Understand the second law of thermodynamics
- Define and calculate entropy
- Applications of 2nd law: engines and refrigerators

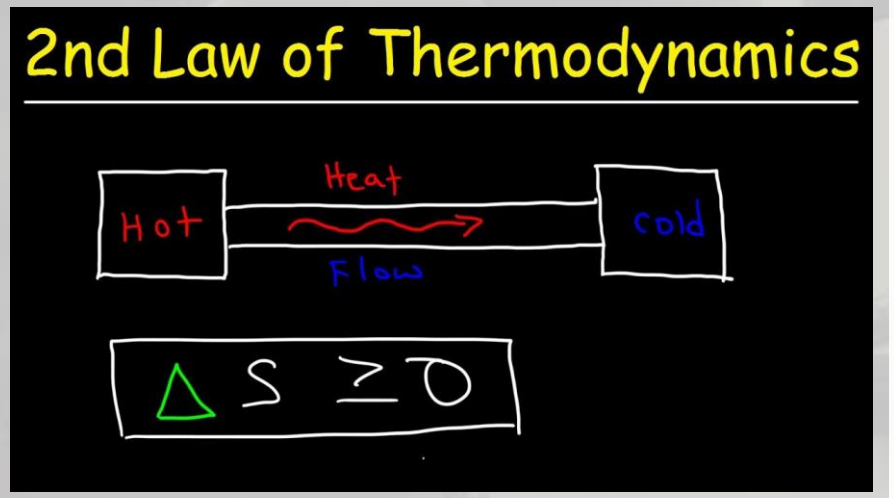
IV. Entropy and the Second Law of Thermodynamics



Introduction

- We are accustomed to many **irreversible process:** An egg is dropped onto a floor, a pizza is baked, a car is driven into a lamppost, etc.
- For reversible processes the system is in equilibrium with its environment, while for irreversible processes it is not.
- Why these processes are irreversible??
- The idea of ***entropy*** provides a mathematical way to encode the intuitive notion of which processes are impossible.

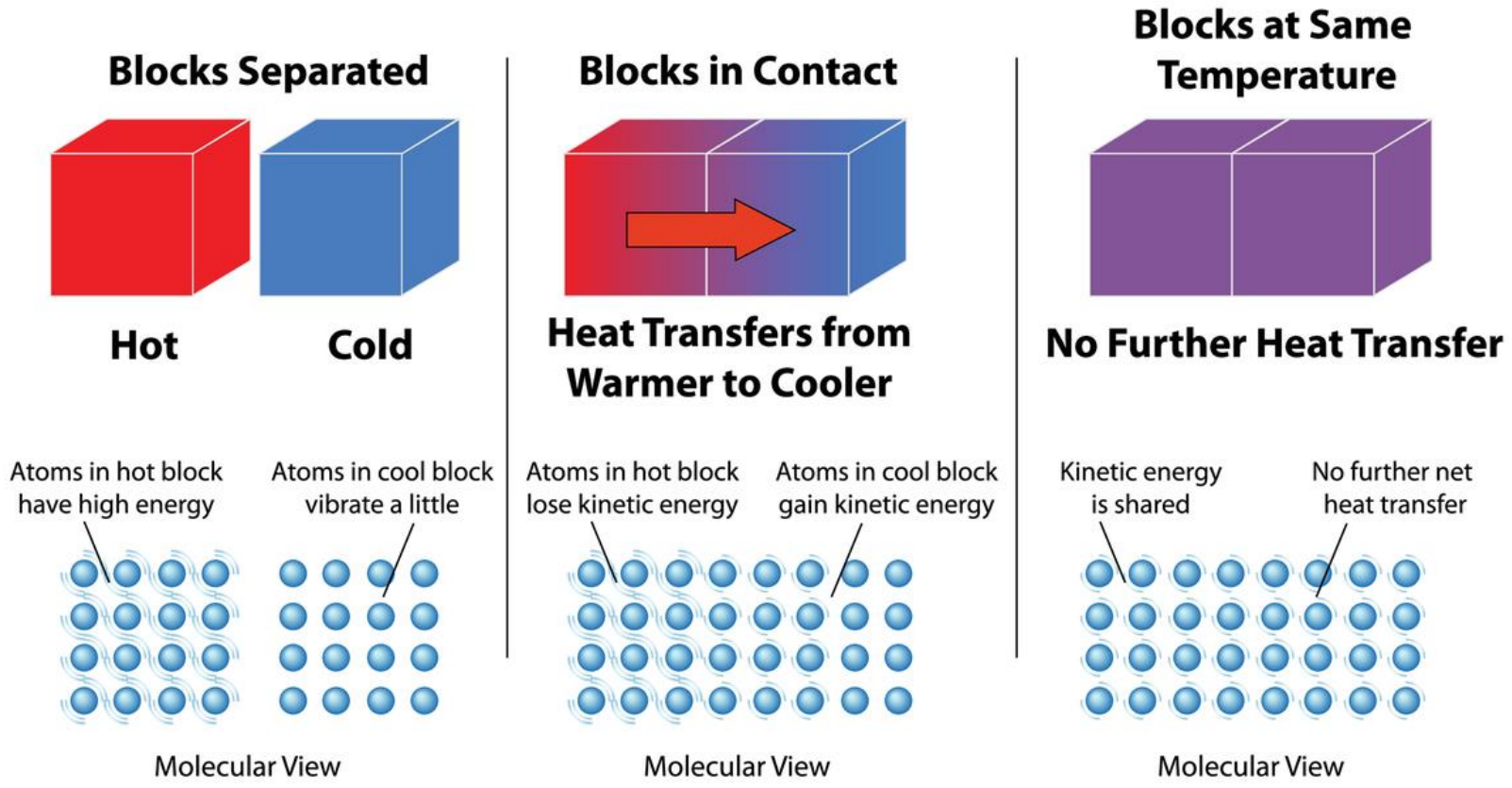
IV. Entropy and the Second Law of Thermodynamics



Introduction

- Entropy: A measure of the molecular "disorder", or randomness, of a system
- The term was coined in 1865 by Clausius from Greek, meaning **transformation, turning, change**
- This concept was first developed in response to the observation that a certain amount of functional energy released from combustion reactions is always lost to dissipation or friction and is thus not transformed into useful work.

Entropy



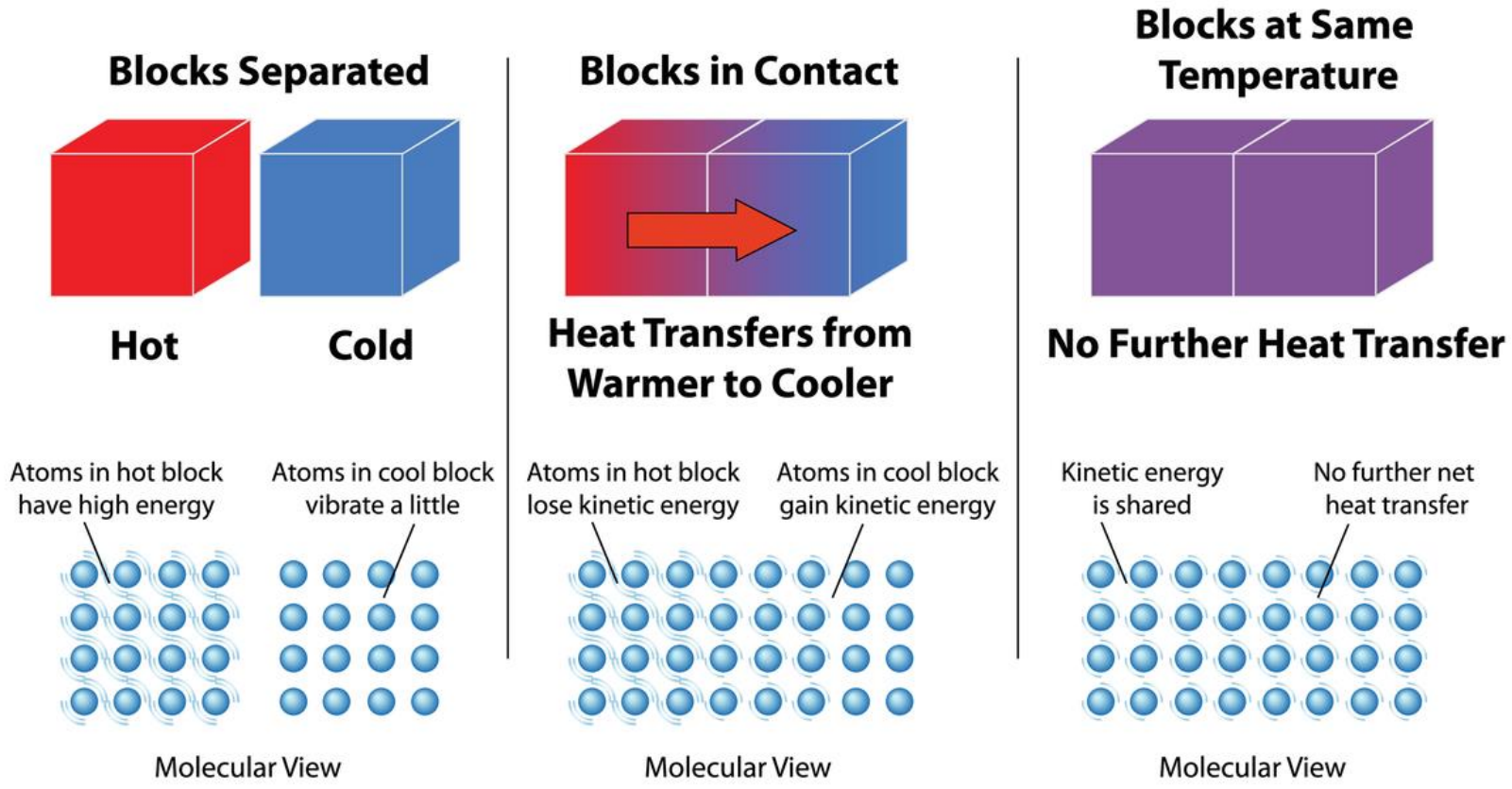
What is entropy?

02 ways to define entropy of a system:

(1) Macroscopic perspective: in terms of the system's temperature and the energy the system gains or loses

(2) Microscopic perspective: by counting the ways in which the atoms or molecules that make up the system can be arranged

Entropy



What is entropy?

02 ways to define entropy of a system:
(1) Macroscopic perspective: in terms of the system's temperature and the energy the system gains or loses

Entropy

Entropy: the measure of a system's thermal energy per unit temperature that is unavailable for doing useful work. Because work is obtained from ordered molecular motion, the amount of entropy is also a measure of the molecular disorder, or randomness, of a system.

Change in entropy: $\Delta S = S_f - S_i = \int_i^f \frac{dQ}{T}$

An entropy change (J/K) depends not only on the energy transferred as heat but also on the temperature at which the transfer takes place

For an isothermal process: $\Delta S = S_f - S_i = \frac{Q}{T}$

When the temperature change is small relative to the temperature before and after the process: $\Delta S = S_f - S_i \approx \frac{Q}{T_{\text{avg}}}$

Entropy is a state property (or a state function): it depends only on those states and not at all on the way the system went from one state to the other.

To find the entropy change for an irreversible process, replace that process with any reversible process that connects the same initial and final states.

The change in entropy is often more useful and practical because it helps us understand how systems evolve during processes, whether a process is spontaneous, and how energy and disorder are distributed. Absolute entropy values are less frequently used in practical applications, except in very specific cases like low-temperature physics.

Entropy as a State Function

Consider a reversible process, with the ideal gas in equilibrium states.

First law in differential form:

$$dE_{\text{int}} = dQ - dW.$$

$$dQ = p\,dV + nC_V\,dT$$

Applying the ideal gas law:

$$\frac{dQ}{T} = nR\frac{dV}{V} + nC_V\frac{dT}{T}$$

$$\int_i^f \frac{dQ}{T} = \int_i^f nR\frac{dV}{V} + \int_i^f nC_V\frac{dT}{T}$$

$$\Delta S = S_f - S_i = nR\ln\frac{V_f}{V_i} + nC_V\ln\frac{T_f}{T_i}$$

Note that we did not have to specify a particular reversible process when we integrated. Therefore, the integration must hold for all reversible processes that take the gas from state i to state f.

The change in entropy between the initial and final states of an ideal gas does not depend on how the gas changes between the two states.

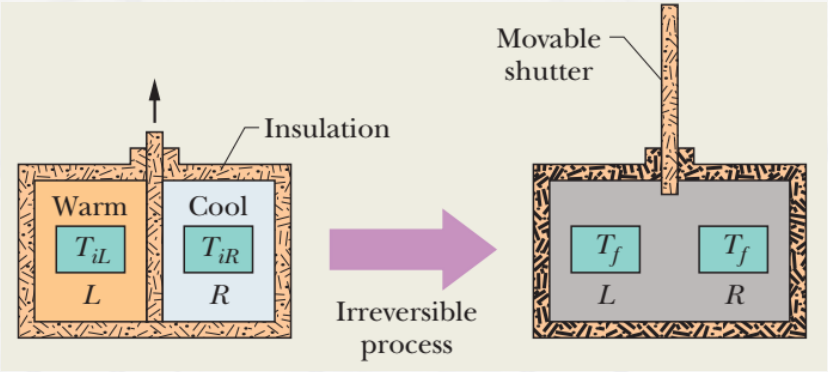
We have assumed that S (like V, P, Eint) is a *state function:* this can be deduced only by experiment.

However, we can prove it is a state function for the special and important case in which an ideal gas is taken through a reversible process.



Exercise

Two identical copper blocks of mass m = 1.5 kg: block L at T_{iL} = 60°C and block R at T_{iR} = 20°C. The blocks are in a thermally insulated box and are separated by an insulating shutter. When we lift the shutter, the blocks eventually come to the equilibrium temperature T_f = 40°C. What is the net entropy change of the two-block system during this irreversible process? The specific heat of copper is 386 J/kg.K





Exercise

$T_{iL} = 60°C$, $T_{iR} = 20°C$, $T_f = 40°C$.

$c_m = 386$ J/kg.K. ΔS?

$$\Delta S_L = \int_i^f \frac{dQ}{T} = \int_{T_{iL}}^{T_f} \frac{mc\,dT}{T} = mc \int_{T_{iL}}^{T_f} \frac{dT}{T}$$

$$= mc \ln \frac{T_f}{T_{iL}}.$$

$$\Delta S_L = (1.5\text{ kg})(386\text{ J/kg}\cdot\text{K}) \ln \frac{313\text{ K}}{333\text{ K}}$$

$$= -35.86\text{ J/K}.$$

$$\Delta S_R = (1.5\text{ kg})(386\text{ J/kg}\cdot\text{K}) \ln \frac{313\text{ K}}{293\text{ K}}$$

$$= +38.23\text{ J/K}.$$

The net entropy change ΔS_{rev} of the two-block system undergoing this two-step reversible process is then

$$\Delta S_{rev} = \Delta S_L + \Delta S_R$$

$$= -35.86\text{ J/K} + 38.23\text{ J/K} = 2.4\text{ J/K}.$$

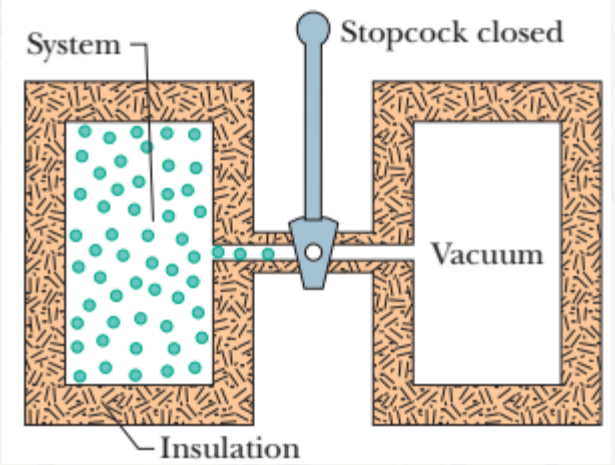
Thus, the net entropy change ΔS_{irrev} for the two-block system undergoing the actual irreversible process is

$$\Delta S_{irrev} = \Delta S_{rev} = 2.4\text{ J/K}. \qquad \text{(Answer)}$$



Exercise

Suppose 1.0 mol of nitrogen gas is confined to the left side of the container. You open the stopcock, and the volume of the gas doubles. What is the entropy change of the gas for this irreversible process? Treat the gas as ideal.





Exercise

$n_{N2} = 1$ mol, $V_2 = 2V_1$. ΔS?

(1) We can determine the entropy change for the irreversible process by calculating it for a reversible process that provides the same change in volume

(2) The temperature of the gas does not change in the free expansion. Thus, the reversible process should be an isothermal expansion

$$Q = nRT \ln \frac{V_f}{V_i},$$

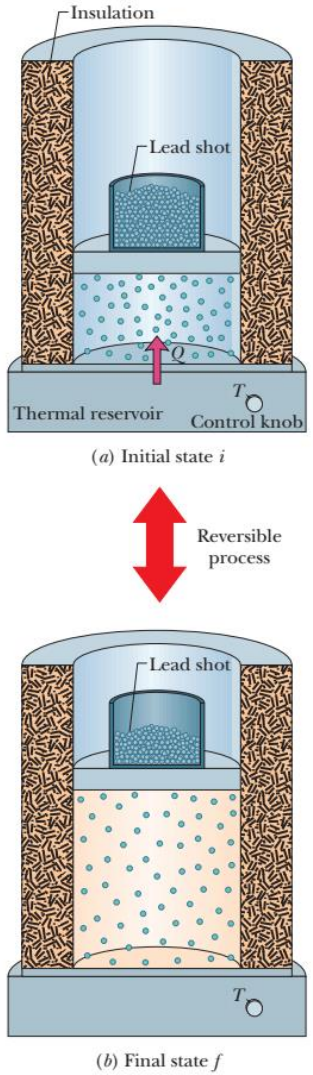
$$\Delta S_{\text{rev}} = \frac{Q}{T} = \frac{nRT \ln(V_f/V_i)}{T} = nR \ln \frac{V_f}{V_i}.$$

$$\Delta S_{\text{rev}} = nR \ln \frac{V_f}{V_i} = (1.00 \text{ mol})(8.31 \text{ J/mol} \cdot \text{K})(\ln 2)$$

$$= +5.76 \text{ J/K}.$$

$$\Delta S_{\text{irrev}} = \Delta S_{\text{rev}} = +5.76 \text{ J/K}.$$

Second law of thermodynamics



The *entropy postulate:*

If a process occurs in a *closed* system, the entropy of the system increases for irreversible processes and remains constant for reversible processes. It never decreases.

The second law of thermodynamics:

$$\Delta S \geq 0$$

(The greater-than sign applies to irreversible processes and the equals sign to reversible processes)

2nd law of thermodynamics

"Many past efforts to provide the Second Law with a firm mathematical foundation. None have succeeded."

But:

"Physics is science based on empirical evidence."

Why is entropy important to study?

There are many reasons, some of them can be:

1. Entropy is central to the Second Law of Thermodynamics that governs the direction of all natural processes and introduces the concept of irreversibility.

It explains why certain processes occur spontaneously (like heat flowing from a hot body to a cold body) while others do not, or why processes like mixing gases or frictional heating cannot be undone without external intervention.

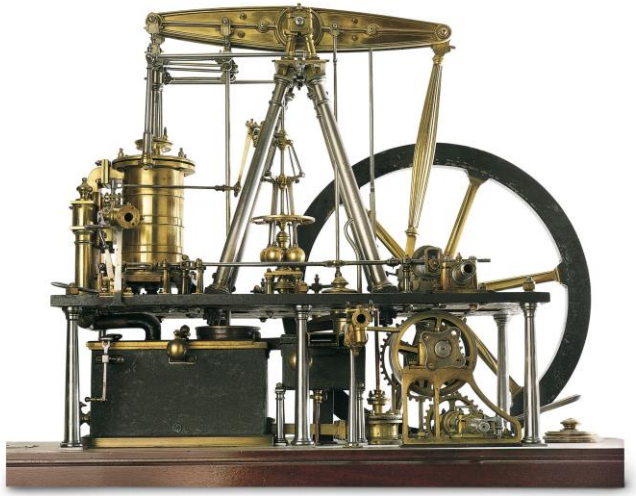
2. Thermal Energy Unavailable for Work

Entropy measures the portion of a system's thermal energy that is not available for doing useful work. The greater the entropy, the less energy is available to do work.

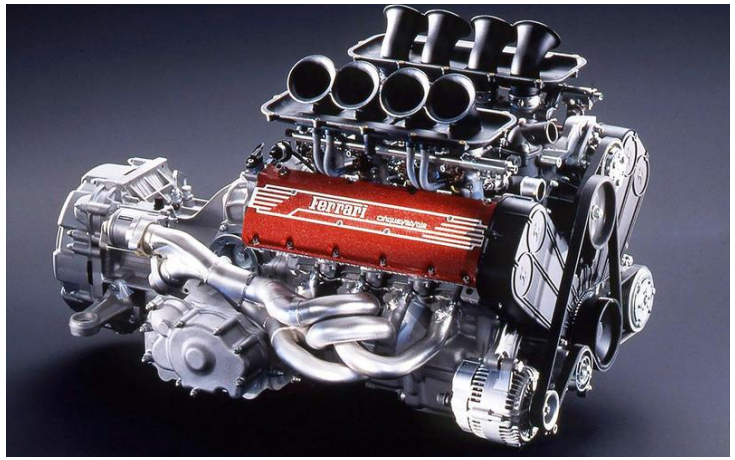
3. Predicting Process Feasibility

Entropy helps determine whether a process is thermodynamically feasible. A process will only occur if the total entropy (system + surroundings) does not decrease.

Entropy in real world: Engines



In a steam engine, the working substance is water, in both its vapor and its liquid form.



In an automobile engine the working substance is a gasoline–air mixture.

Heat engines

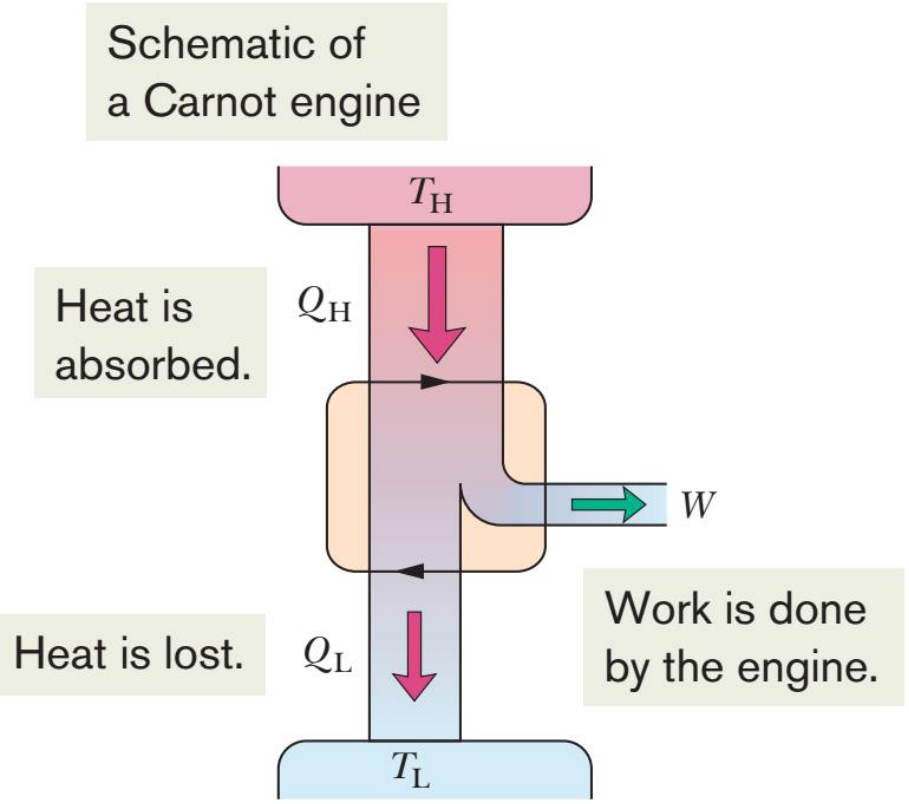
A **heat engine** (more simply, an **engine**)**:** a device that extracts energy from its environment in the form of heat and does work.

At the heart of every engine is a *working substance*.

If an engine is to do work on a sustained basis, the working substance must operate in a *cycle*.

The working substance must pass through a closed series of thermodynamic processes, called *strokes*, returning again and again to each state in its cycle.

Entropy in real world: Engines



The two black arrowheads on the central loop suggest the working substance operating in a cycle, as if on a p-V plot.

$|Q_H|$ + is transferred as heat from the high-temperature reservoir at T_H to the working substance.

$|Q_L|$ is transferred as heat from the working substance to the low-temperature reservoir at T_L.

W is done by the engine (actually by the working substance) on something in the environment.

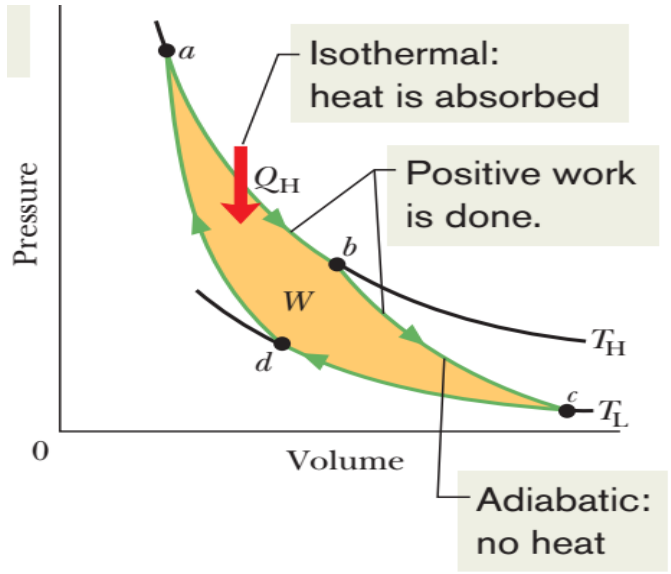
Carnot engine

In an ideal engine, all processes are reversible and no wasteful energy transfers occur due to, say, friction and turbulence.

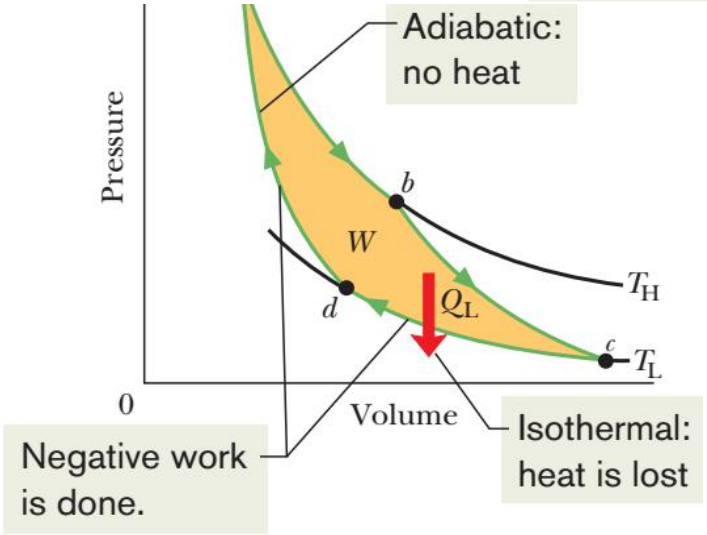
Let's focus on a particular ideal engine called a **Carnot engine**: the best (in principle) at using energy as heat to do useful work.

Carnot was able to analyze the performance of this engine before the first law of thermodynamics and the concept of entropy had been discovered!!!

Entropy in real world: Engines



If we place the cylinder in contact with the high temperature reservoir at T_H, $|Q_H|$ is transferred *to* the working substance *from* this reservoir as the gas undergoes an isothermal *expansion* from V_a to V_b.



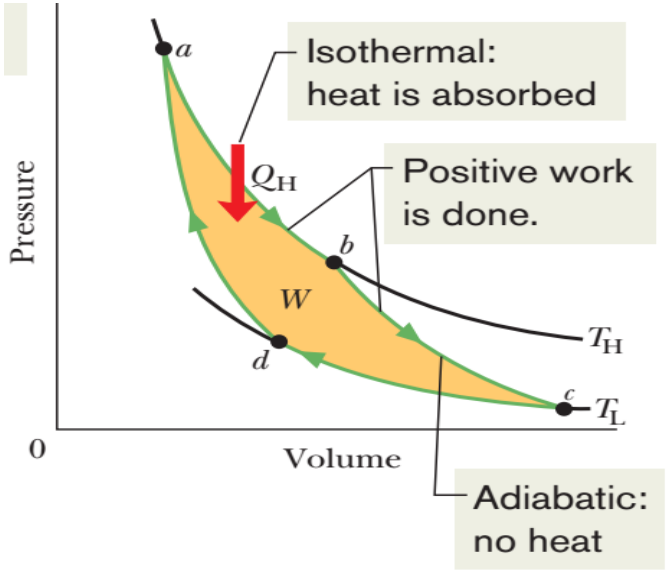
With the working substance in contact with the low-temperature reservoir at T_L, $|Q_L|$ is transferred *from* the working substance *to* the low-temperature reservoir as the gas undergoes an isothermal *compression* from V_c to V_d

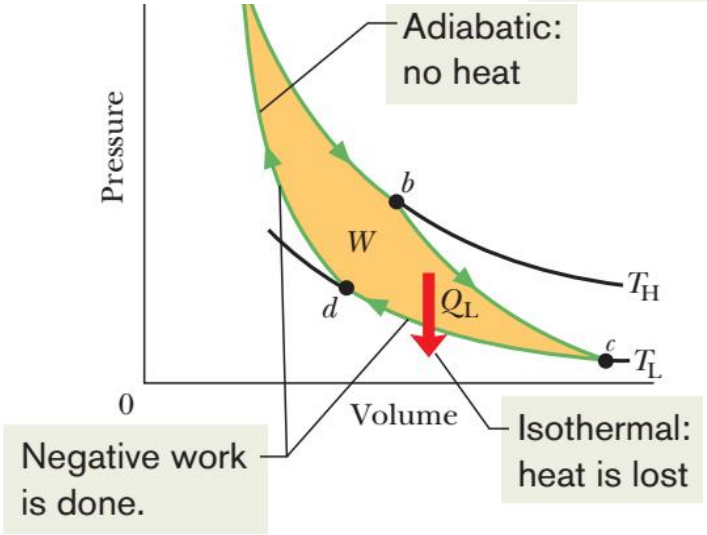
Carnot engine

Imagine the working substance to be a gas, confined to an insulating cylinder with a weighted, movable piston.

The cylinder may be placed at will on either of the two thermal reservoir.

Entropy in real world: Engines





Assuming that heat transfers to or from the working substance can take place *only* during the isothermal processes *ab* and *cd*.

Therefore, processes *bc* and *da* which connect the two isotherms must be (reversible) adiabatic processes.

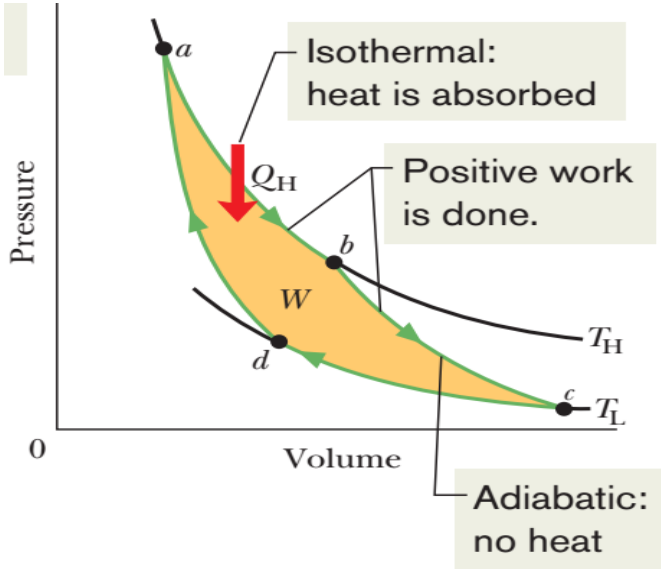
To ensure this, during processes *bc* and *da* the cylinder is placed on an insulating slab as the volume of the working substance is changed.

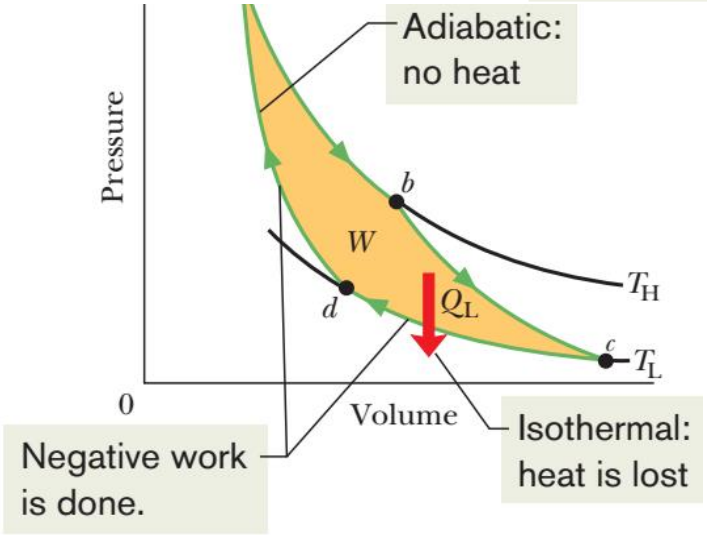
Carnot engine

Imagine the working substance to be a gas, confined to an insulating cylinder with a weighted, movable piston.

The cylinder may be placed at will on either of the two thermal reservoir.

Entropy in real world: Engines





Work done by a Carnot engine:

$$W = |Q_H| - |Q_L|.$$

Entropy changes:

$$\Delta S = \Delta S_H + \Delta S_L = \frac{|Q_H|}{T_H} - \frac{|Q_L|}{T_L}$$

Because entropy is a state function, $\Delta S = 0$ for a complete cycle:

$$\frac{|Q_H|}{T_H} = \frac{|Q_L|}{T_L}$$

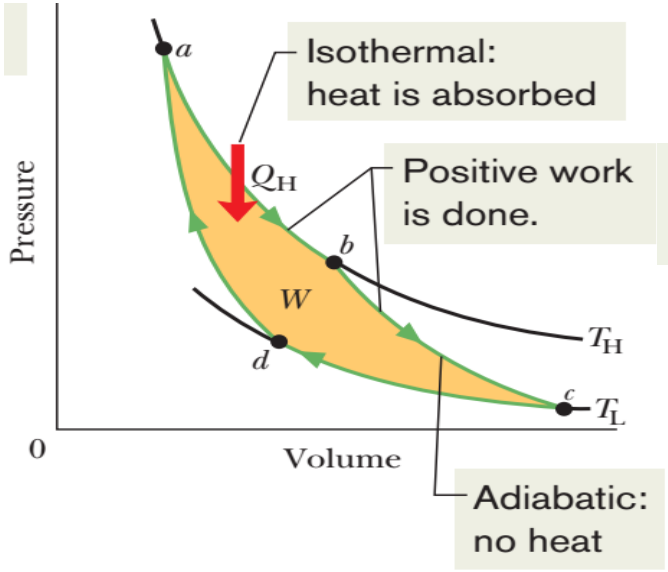
Because $T_H > T_L$, $|Q_H| > |Q_L|$: more energy is extracted as heat from the high-temperature reservoir than is delivered to the low-temperature reservoir.

Carnot engine

Work done by a Carnot engine: for a closed cycle: $W_{net} = Q_{net}$ *(first law of thermodynamics)*

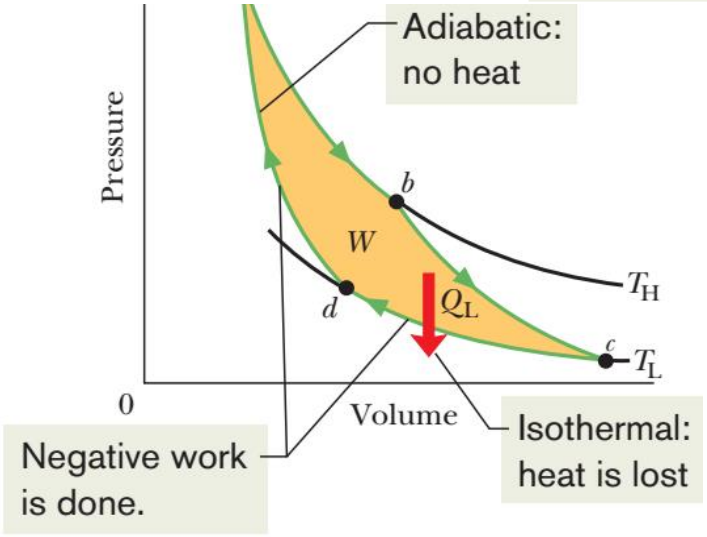
Entropy changes: only two reversible energy transfers as heat, and thus two changes in the entropy of the working substance, one at temperature T_H and one at T_L.

Entropy in real world: Engines



$$\varepsilon = \frac{\text{energy we get}}{\text{energy we pay for}} = \frac{|W|}{|Q_H|} \quad \text{(efficiency, any engine)}$$

$$\varepsilon_C = \frac{|Q_H| - |Q_L|}{Q_H} = 1 - \frac{|Q_L|}{|Q_H|}$$

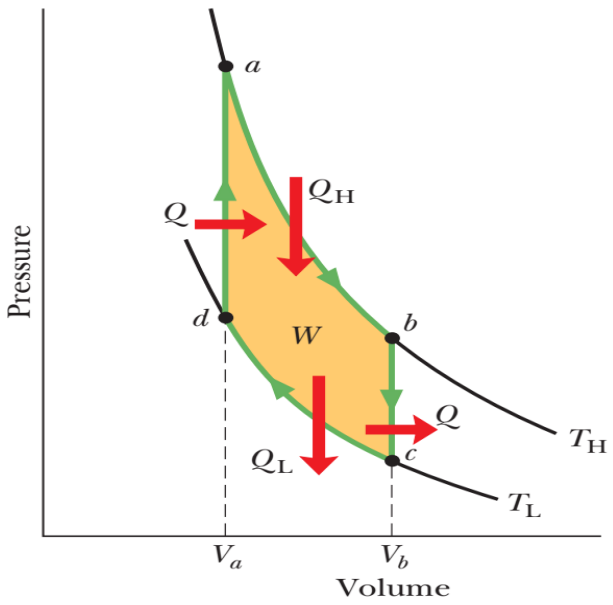


$$\varepsilon_C = 1 - \frac{T_L}{T_H} \quad \text{(efficiency, Carnot engine)}$$

Carnot engine

No real engine can have a thermal efficiency greater than Carnot engine!

Entropy in real world: Engines



A p-V plot for the working substance of an ideal Stirling engine

Isothermal heat transfers at temperatures T_H and T_L

Transfers of energy as heat Q is required to increase/decrease the temperature between, T_L and T_H

$$\varepsilon_S = \frac{|W|}{|Q'_H|} = \frac{(Q_H+Q)-(Q_L+Q)}{Q_H+Q} = \frac{Q_H-Q_L}{Q_H+Q} < \frac{Q_H-Q_L}{Q_H} = \varepsilon_C$$

Stirling engine

The Stirling engine was developed in 1816 by Robert Stirling.

Applications:
- automobiles
- spacecraft
- military submarines

(because these engines are more quiet than others)



Exercise

Imagine a Carnot engine that operates between the temperatures T_H = 850 K and T_L = 300 K. The engine performs 1200 J of work each cycle, which takes 0.25 s.
(a) What is the efficiency of this engine?

(b) What is the average power of this engine?

(c) How much energy $|Q_H|$ is extracted as heat from the high-temperature reservoir every cycle?

(d) How much energy $|Q_L|$ is delivered as heat to the low temperature reservoir every cycle?

(e) By how much does the entropy of the working substance change as a result of the energy transferred to it from the high-temperature reservoir? From it to the low-temperature reservoir?



Exercise

T_H = 850 K and T_L = 300 K. W = 1200 J each cycle of 0.25 s.

(a) What is the efficiency of this engine?

$$\varepsilon = 1 - \frac{T_L}{T_H} = 1 - \frac{300\ \text{K}}{850\ \text{K}} = 0.647 \approx 65\%$$

(b) What is the average power of this engine?

$$P = \frac{W}{t} = \frac{1200\ \text{J}}{0.25\ \text{s}} = 4800\ \text{W} = 4.8\ \text{kW}$$

(c) How much energy $|Q_H|$ is extracted as heat from the high-temperature reservoir every cycle?

$$|Q_H| = \frac{W}{\varepsilon} = \frac{1200\ \text{J}}{0.647} = 1855\ \text{J}$$

(d) How much energy $|Q_L|$ is delivered as heat to the low temperature reservoir every cycle?

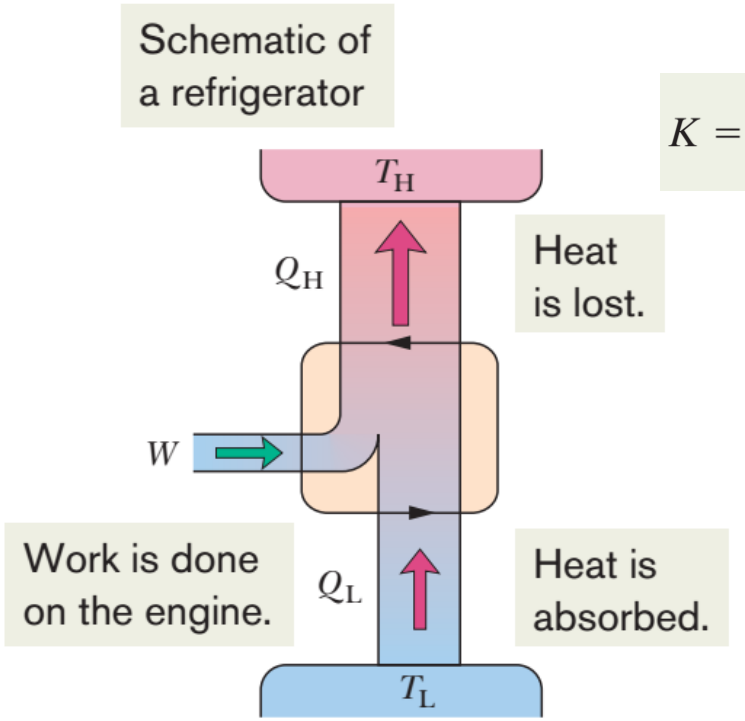
$$|Q_L| = |Q_H| - W$$

$$= 1855\ \text{J} - 1200\ \text{J} = 655\ \text{J}$$

(e) By how much does the entropy of the working substance change as a result of the energy transferred to it from the high-temperature reservoir? From it to the low-temperature reservoir?

$$\Delta S_H = \frac{Q_H}{T_H} = \frac{1855\ \text{J}}{850\ \text{K}} = +2.18\ \text{J/K} \qquad \Delta S_L = \frac{Q_L}{T_L} = \frac{-655\ \text{J}}{300\ \text{K}} = -2.18\ \text{J/K}$$

Entropy in real world: Refrigerators



Coefficient of performance

$$K = \frac{\text{what we want}}{\text{what we pay for}} = \frac{|Q_L|}{|W|}$$ (coefficient of performance, any refrigerator),

$$K_C = \frac{|Q_L|}{|Q_H| - |Q_L|}.$$

$$K_C = \frac{T_L}{T_H - T_L}$$ (coefficient of performance, Carnot refrigerator).

Refrigerator

A **refrigerator:** device that uses work to transfer energy from a low temperature reservoir to a high-temperature reservoir

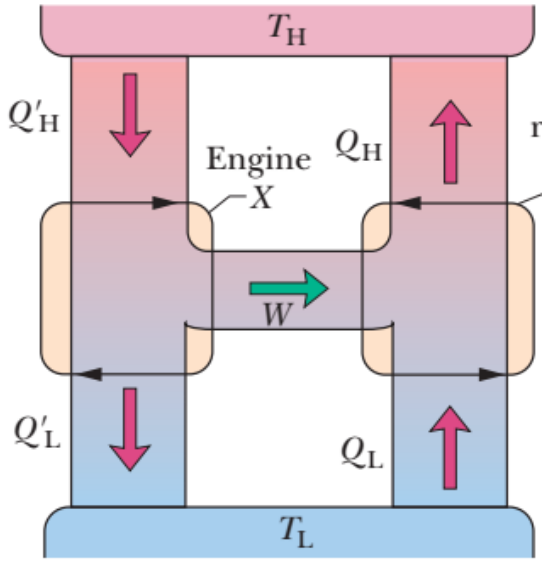
In an ideal refrigerator, all processes are reversible and no wasteful energy transfers occur as a result of, say, friction and turbulence.

Its operation is the reverse of how the Carnot engine operates.

Efficiency of real engines:

Assume an engine X h $\varepsilon_X > \varepsilon_C$ (a claim)

Let's couple engine *X* to a Carnot refrigerator.



(*a*)

Carnot refrigerator's stroke is adjusted so that the work it requires per cycle is just equal to that provided by engine *X* ($W_C = W_X$).

Thus, no (external) work is performed on or by the system *engine* + *refrigerator*

Engine efficiency

Here we prove that no real engine can have an efficiency greater than that of a Carnot engine.

If it could, the engine would violate the second law of thermodynamics

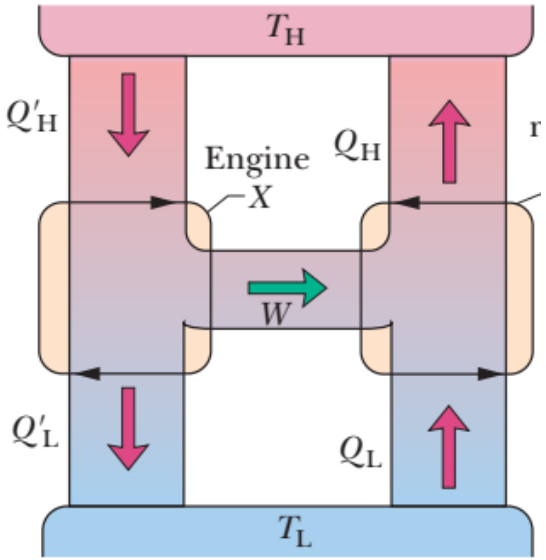
Efficiency of real engines:

If $\varepsilon_X > \varepsilon_C$ is true:

$$\frac{|W|}{|Q'_H|} > \frac{|W|}{|Q_H|}$$

Then:

$$|Q_H| > |Q'_H|$$



(a)

Because the $W_X = W_C$, from the first law of thermodynamics:

$$|Q_H| - |Q_L| = |Q'_H| - |Q'_L| \quad \text{or}$$

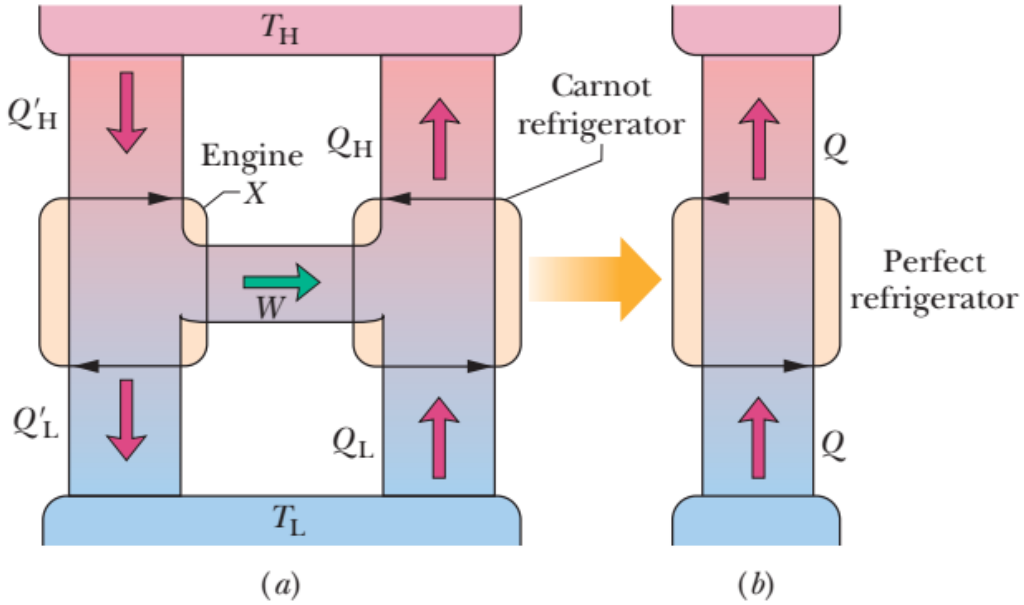
$$|Q_H| - |Q'_H| = |Q_L| - |Q'_L| = Q > 0$$

Engine efficiency

Here we prove that no real engine operating between can have an efficiency greater than that of a Carnot engine.

If it could, the engine would violate the second law of thermodynamics

Efficiency of real engines:



The transfer energy Q as heat from a low to a high-temperature reservoir without the requirement of work → perfect refrigerator!!!

A perfect refrigerator transfers energy as heat Q from a cold reservoir to a warm reservoir without the need for work.

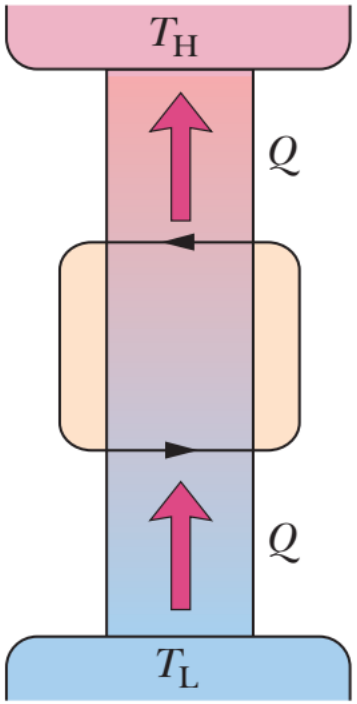
Engine efficiency

Here we prove that no real engine operating between can have an efficiency greater than that of a Carnot engine.

If it could, the engine would violate the second law of thermodynamics

Efficiency of real engines:

Perfect refrigerator: total transfer of heat from cold to hot without any work



Because the unit operates in cycles, the entropy of the working substance does not change during a complete cycle.

The entropies of the two reservoirs, however, do change: $\Delta S_L = -|Q|/T$ and $\Delta S_H = +|Q|/T$

Thus, the net entropy change for the entire system is

$$\Delta S = -\frac{|Q|}{T_L} + \frac{|Q|}{T_H}$$

Because $T_H > T_L$ → $\Delta S < 0$.

Violation of the second law → a perfect refrigerator does not exist!!

Engine efficiency

No real engine can have an efficiency greater than that of a Carnot engine when both engines work between the same two temperatures

Exercise

1. Which of the following is true?
a) for an isolated system, $dS>=0$
b) for a reversible process, $dS=0$
c) for an irreversible process, $dS>0$
d) all of the mentioned

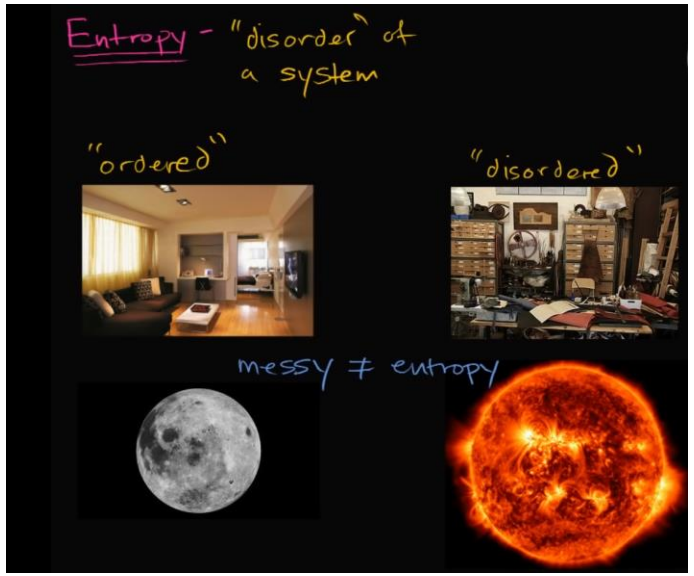
Answer: d). For an isolated system which does not undergo any energy interaction with the surroundings, $dQ=0$ and also $dS>=dQ/T$.

2. Entropy may decrease locally at some region within the isolated system. How can this statement be justified?
a) this cannot be possible
b) this is possible because entropy of an isolated system can decrease.
c) it must be compensated by a greater increase of entropy somewhere within the system.
d) none of the mentioned

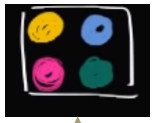
Answer: c). The net effect of an irreversible process is an entropy increase of the whole system.

Entropy

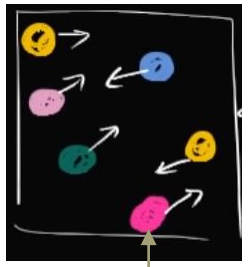
The « disorder of » entropy is of the number of states that a system can take on



Consider 2 systems: one with 4 atoms, one with eight atoms



Fewer possible states



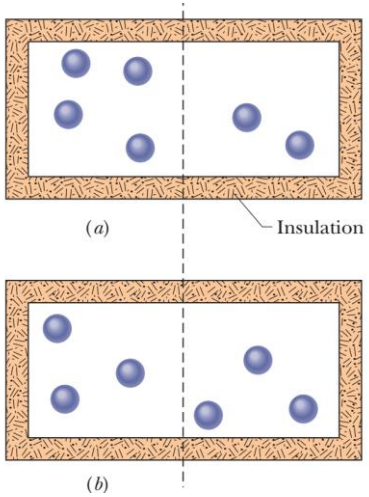
More possible states
More entropy!

Ludwig Boltzmann defined entropy as a measure of the number of possible microscopic states (*microstates*) of a system in thermodynamic equilibrium

What is entropy?

02 ways to define entropy of a system:
(2) microscopic perspective: by counting the ways in which the atoms or molecules that make up the system can be arranged

Entropy



Consider a box that contains six identical (and thus indistinguishable) molecules of a gas

At any instant, a given molecule will be in either the left or the right half (having the same volume) of the box: the molecule has the same probability of being in either half.

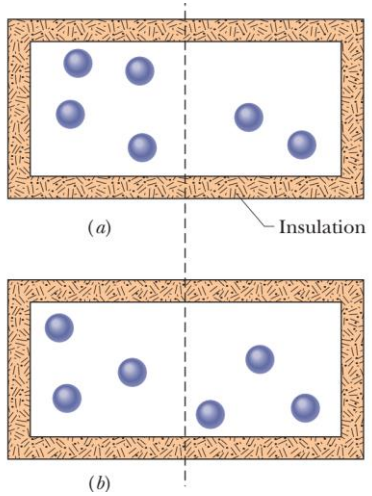
Possible *configurations* of the six molecules:

Configuration Label	n_1	n_2
I	6	0
II	5	1
III	4	2
IV	3	3
V	2	4
VI	1	5
VII	0	6

What is entropy?

02 ways to define entropy of a system:
(1) microscopic perspective: by counting the ways in which the atoms or molecules that make up the system can be arranged

Entropy



Configuration		
Label	n_1	n_2
I	6	0
II	5	1
III	4	2
IV	3	3
V	2	4
VI	1	5
VII	0	6

A given configuration can be achieved in a number of different ways. We call these different arrangements of the molecules *microstates*.

Suppose N molecules are distributed with n_1 molecules in one half of the box and n_2 in the other ($n_1 + n_2 = N$)

The total number of ways in which we can select all six molecules is the product of these independent ways,
or 6 x 5 x 4 x 3 x 2 x 1 = 720 or 6! = 720

What is entropy?

02 ways to define entropy of a system:
(1) microscopic perspective: by counting the ways in which the atoms or molecules that make up the system can be arranged

Entropy

Configuration			Multiplicity W (number of microstates)	Calculation of W
Label	n_1	n_2		
I	6	0	1	6!/(6! 0!) = 1
II	5	1	6	6!/(5! 1!) = 6
III	4	2	15	6!/(4! 2!) = 15
IV	3	3	20	6!/(3! 3!) = 20
V	2	4	15	6!/(2! 4!) = 15
VI	1	5	6	6!/(1! 5!) = 6
VII	0	6	1	6!/(0! 6!) = 1
			Total = 64	

However, because the molecules are indistinguishable, these 720 arrangements are not all different.

To get the number of *different* arrangements:

$$W = \frac{N!}{n_1!\, n_2!} \quad \text{(multiplicity of configuration)}$$

(the number of microstates that correspond to a given configuration)

What is entropy?

02 ways to define entropy of a system:
(1) microscopic perspective: by counting the ways in which the atoms or molecules that make up the system can be arranged

Entropy

Configuration			Multiplicity W (number of microstates)	Calculation of W
Label	n_1	n_2		
I	6	0	1	6!/(6! 0!) = 1
II	5	1	6	6!/(5! 1!) = 6
III	4	2	15	6!/(4! 2!) = 15
IV	3	3	20	6!/(3! 3!) = 20
V	2	4	15	6!/(2! 4!) = 15
VI	1	5	6	6!/(1! 5!) = 6
VII	0	6	1	6!/(0! 6!) = 1
			Total = 64	

Configuration IV: *most probable configuration,* with probability 20/64 = 0.313 (system is in configuration IV 31.3% of the time)

→ The molecules are evenly divided between the two halves of the box, because that is what we expect at thermal equilibrium.

Configurations I and VII, (all the molecules are in one half of the box) are the least probable (probability 1/64 = 0.016 or 1.6%)

→ There is any probability, however small, of finding all six molecules clustered in half of the box, with the other half empty

For large values of N: nearly all the microstates belong to the configuration in which the molecules are divided equally between the two halves of the box, and probability for configurations I an VII are **mostly 0**!!

What is entropy?

02 ways to define entropy of a system:

(1) microscopic perspective: by counting the ways in which the atoms or molecules that make up the system can be arranged

Entropy

Configuration Label	n_1	n_2	Multiplicity W (number of microstates)	Calculation of W	Entropy 10^{-23} J/K
I	6	0	1	6!/(6! 0!) = 1	0
II	5	1	6	6!/(5! 1!) = 6	2.47
III	4	2	15	6!/(4! 2!) = 15	3.74
IV	3	3	20	6!/(3! 3!) = 20	4.13
V	2	4	15	6!/(2! 4!) = 15	3.74
VI	1	5	6	6!/(1! 5!) = 6	2.47
VII	0	6	1	6!/(0! 6!) = 1	0
			Total = 64		

In 1877, Boltzmann derived a relationship between the entropy *S* of a configuration of a gas and the multiplicity *W* of that configuration:

$$S = k \ln W \quad \text{(Boltzmann's entropy equation)}$$

What is entropy?

02 ways to define entropy of a system:
(1) microscopic perspective: by counting the ways in which the atoms or molecules that make up the system can be arranged



Exercise

Suppose that there are 100 indistinguishable molecules in the previous box. How many microstates are associated with the configuration $n1 = 50$ and $n2 = 50$, and with the configuration $n1 = 100$ and $n2 = 0$? Interpret the results in terms of the relative probabilities of the two configurations



Exercise

How many microstates are associated with the configuration $n1 = 50$ and $n2 = 50$, and with the configuration $n1 = 100$ and $n2 = 0$?

Configuration (50, 50):

$$W = \frac{N!}{n_1!\, n_2!} = \frac{100!}{50!\, 50!}$$

$$= \frac{9.33 \times 10^{157}}{(3.04 \times 10^{64})(3.04 \times 10^{64})}$$

$$= 1.01 \times 10^{29}.$$

Configuration (100, 0):

$$W = \frac{N!}{n_1!\, n_2!} = \frac{100!}{100!\, 0!} = \frac{1}{0!} = \frac{1}{1} = 1$$

A 50 – 50 distribution is more likely than a 100 – 0 distribution by the factor of about 10^{29}

Summary

- Reversible and Irreversible process
- Entropy: two ways to define
- Change in Entropy
- The second law of thermodynamics
- Engines
- Refrigerators